# Yasaman Moradi Fard

**Research Interests:** Medical Image Analysis · Deep Learning for Clinical Imaging . Affective Computing · LLMs for Clinical Text Analysis

## Personal


- **Date of Birth:** 30. November.1996
- **Address:** Theaterplatz 18, 91054 Erlangen
- **Mobile:** 015751181777
- **E-mail:** yasaman.moradifard@gmail.com
- **LinkedIn:** Yasaman Moradi


## Education

- **Friedrich-Alexander-Universität, Erlangen, Germany** 2021 - 2026
  M.Sc. Medizintechnik
  Master-Thesis: *Multimodal Depression Detection & Monitoring*
  - End-to-End-Pipeline for Multi-modal Depression Monitoring with 93% F1-weighted
  - Benchmarking from ML/DL/Transformer-models including Multi-modal Fusions
  - Llama 3.3 & DepRoBERTa for text-based feature extraction
  - Dataset: Empkins-Ekspression (Link to dataset)

- **University of Isfahan, Isfahan, Iran** 2015 - 2019
  B.Sc. Medical Engineering (Bio-Electric)
  Bachelor-Thesis: *Neural Decoding*
  - Clustering und classification to detect statistical patterns in EEG Signals

## Working Experience

- **Research Assistant, Machine Learning and Data Analysis Lab.** Nov 2024 - Jul 2025
  Erlangen, Germany – Supervised by: Misha Sadeghi & Prof. Björn Eskofier
  - Data collection and machine learning for depression screening.
  - Signal Processing & Data mining on ECG, RSP, EMG, video, audio, and text datasets.

- **Intern at Medical Image and Signal Processing Research Center** Jun-Sep 2018
  Isfahan University of Medical Science, Isfahan, Iran
  - Contributing to an image processing project including lung nodules detection
  - Applying filters to image data preprocessing
  - Applying CNN model on lung CT images
  - Applying Python programming and statistical methods

- **Working Student, Siemens Healthineers** 2022 - 2024
  Erlangen, Germany
  - Analyzed clinical product performance data in a medical device context
  - Dashboard reporting on sales data (e.g., Power BI / Python).

- **Talent Programm, Zollhof Tech-Incubator** Jul - Sep 2025
  Nürnberg, Germany
  - Conducted user interviews, analyzed pain points and workflows, and documented requirements.
  - Developed solution concepts for user needs, with a focus on data consolidation and data quality as the foundation for automation.
  - Built a human-in-the-loop prototype for AI-assisted workflow automation

## Research Experience

- **Advanced Deep Learning Project in Medical Image Augmentation** Apr - Sep 2024
  Pattern Recognition Lab., Erlangen, Germany
  - Further development of advanced deep learning methods for medical image augmentation
  - Data Augmentation in Medical Image Analysis in Segmentation and Classification tasks

- **Research collaboration, King Abdullah University of Science and Technology** May - Nov 2020
  Saudi Arabia (Remote)
  - Y. Moradi Fard, et al. "Deep learning in the industrial internet of things: Potentials, challenges, and emerging applications." *IEEE Internet of Things Journal* 8.14 (2021): 11016-11040 (Link to Paper)

## Skills

- **ML & AI:** PyTorch, TensorFlow, scikit-image, scikit-learn, XGBoost, LightGBM, CNN, Transfer Learning, Transformers, Multimodal Fusion strategies, Affective Computing
- **Generative AI & LLMs:** LLAMA 3.3, DepRoBERTa, Prompt Engineering, Hugging Face Transformers, RAG
- **Signal Processing:** OpenCV, OpenSMILE, OpenFace, OpenDBM, ECG/EMG/RSP analysis, HRV analysis, MFCC/spectral features, Facial Action Units
- **Programming:** Python, SQL, C++, Git/GitLab, Linux, Docker
- **Data & Research:** pandas, SciPy, NumPy, matplotlib, seaborn, Power BI, Statistical Analysis, Data Mining, Feature Engineering, Experimental Design, Scientific Writing
- **Language:** English (fluent), German (B2), Persian (Native)

## Volunteer

- **START Nürnberg** Since 2023
  Nürnberg, Germany
  - Networking and community building for the next generation of founders

## Workshops & Relevant Courses

- **Medical Device Regulation (MDR),** FAU 2024
- **Business and Startup Evaluation,** Digital Tech Academy, FAU 2022
- **Advanced Statistical Methods for Scientific Reports,** University of Isfahan 2019
- **Medical Data Mining,** University of Isfahan 2018
- **EMG Signal Processing & Data Mining,** University of Isfahan 2018